UC IPM

< Corn

AGRICULTURE

Aphids

Corn leaf aphid: *Rhopalosiphum maidis*

Green peach aphid: *Myzus persicae*

Greenbug: *Schizaphis graminum*

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Description of the Pest

Key to identifying aphids

Several species of aphids may be found in corn, but corn leaf aphid and greenbug are the primary aphid species infesting corn in California. Corn leaf aphids are small to medium and bluish green in color and also infest small grains. The greenbug is a moderate-sized aphid. The color of the abdomen is light green with a **darker stripe** down the middle. Both winged and wingless forms of both aphids occur on corn plants.

Damage

Corn leaf aphid infestations usually start in the plant whorl. Heavy infestations may curl leaves and stunt the plant. Later infestations may completely cover the tassels and upper leaves. Corn leaf aphids excrete a sticky substance called honeydew, which accumulates on the plants. The honeydew eventually turns blackish as sooty molds grow on it. Heavy amounts of sooty mold may be more damaging to silage corn than to corn for grain.

Greenbugs and **green peach aphids** also infest corn, but usually do not build up to the high numbers of corn leaf aphids. Red lesions often form at the feeding sites of greenbugs. High numbers of greenbugs on small plants can kill the plants. All three species transmit maize dwarf mosaic virus to corn from nearby sources. **Johnsongrass** is one of the common weed hosts for this virus.

Management

Transmission of virus disease is the primary damage caused by aphids and the potential for this varies significantly from year-to-year and area-to-area. Insecticide sprays will not prevent virus transmission, but can reduce population levels. There are no established thresholds for aphids on field corn. Only on rare occasions do aphids reach damaging populations. Obtaining good coverage of the plant, which is essential for effective control, can be difficult when the plants are 5 feet tall or more and treatments may increase problems with mites by killing natural enemies.

Biological Control

Aphids can be kept below economic levels of feeding damage by the parasite *Lysiphlebus testaceipes* and by predators such as lacewings, lady beetles, and syrphid flies. However, biological control cannot prevent transmission of virus diseases.

Organically Acceptable Methods

Biological control and oil and soap sprays are acceptable for use on organically grown crops.

Pesticides and Natural Enemy Releases

Not all registered pesticides are listed. **The following are ranked by their IPM value, with the most effective and least harmful to natural enemies, honey bees, and the environment listed at the top of the table.** When choosing a pesticide, consider information related to water and air quality, resistance management, and the pesticide's properties and application timing. Use PestManage to compare summarized management options for different pests in the same crop. Always carefully read the label of the product being used and take all necessary precautions when handling pesticides.

Expand table

Rank	Active ingredient	Example trade name
Application Timing Varies (See *UC IPM Pest Manageme*		
A	dimethoate	Dimethoate 400
B	esfenvalerate	Asana XL

Rank	Active ingredient	Example trade name
C	endosulfan *	Thionex 3EC
D	mineral oil	various products #

—	No information.
I	Do not apply or allow to drift to plants that are flowering including weeds. Do not allow pesticide to contaminate water accessible to bees including puddles.
II	Do not apply or allow to drift to plants that are flowering including weeds, except when the application is made between sunset and midnight if allowed by the pesticide label and regulations. Do not allow pesticide to contaminate water accessible to bees including puddles.
III	No bee precaution, except when required by the pesticide label or regulations.

- ‡ a b Restricted entry interval (REI) is the number of hours (unless otherwise noted) from treatment until the treated area can be safely entered without personal protective equipment. Preharvest interval (PHI) is the number of days from treatment to harvest. In some cases, the REI exceeds the PHI. The longer of the two intervals is the minimum time that must elapse before harvest.
- * a Permit required from county agricultural commissioner for purchase or use.
- # ↩ Acceptable for use on certified organic crops. Check with your certifier to confirm before application.
- [1] ↩ Group numbers for insecticides and miticides are assigned by the Insecticide Resistance Action Committee (IRAC). Insecticides with unknown modes of action are assigned mode-of-action group

numbers (MoAs) that begin with UN. Rotate pesticides with a different mode-of-action group number, and do not use products with the same mode-of-action group number more than twice per season to help prevent the development of resistance. For example, the organophosphates have a group number of 1B; insecticides with a 1B group number should be alternated with insecticides that have a group number other than 1B.

- [2] ↩ Range of insect and mite groups affected by a pesticide. **Broad** means the pesticide affects most groups of insects and mites; **narrow** means the pesticide affects only a few specific groups.
- [3] ↩ Risk of harm to honey bees. For more information, see ***Bee Precaution Pesticide Ratings***.
- [4] ↩ Risk of harm to predatory mites. Toxicities are generally to western predatory mite, *Galendromus occidentalis*. Where differences have been measured in toxicity of the pesticide-resistant strain versus the native strain, these are listed as pesticide-resistant strain or native strain.
- [5] a b Risk of harm to parasitoids and general predators. Toxicities are averages of reported effects and should be used only as a general guide. Actual toxicity of a specific insecticide depends on factors including the application rate, environmental conditions, and the life stage and species of a parasitoid or predator.
- [6] ↩ Length of time residue affects natural enemies. **Short** means hours to days; **moderate** means days to 2 weeks; and **long** means many weeks or months.
- [7] ↩ Risk of harm to fish from leaching, based on USDA Natural Resources Conservation Service Windows Pesticide Screening Tool (WIN-PST).
- [8] ↩ Risk of harm to fish from adsorbed runoff, based on USDA Natural Resources Conservation Service Windows Pesticide Screening Tool (WIN-PST).

- 9 ↩ Risk of harm to fish from solution runoff, based on USDA Natural Resources Conservation Service Windows Pesticide Screening Tool (WIN-PST).
- 10 ↩ Risk of harm to humans from leaching, based on USDA Natural Resources Conservation Service Windows Pesticide Screening Tool (WIN-PST).
- 11 ↩ Risk of harm to humans from solution runoff, based on USDA Natural Resources Conservation Service Windows Pesticide Screening Tool (WIN-PST).
- 12 ↩ Date information was last updated in the *UC IPM Pest Management Guidelines*.

UC PEER REVIEWED

UC IPM Pest Management Guidelines: Corn

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